

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), and Text2Cypher (Ozsoy et al., 2025), show that

high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery (Zheng et al., 2024) motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

3 Method

PIPE-Cypher has five stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A

lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5 models for generation and judging.

The intended full-quality model is Qwen3.5-35B-A3B, but our June 1, 2026 live evidence uses the cached Qwen3.5-9B fallback. The 35B-A3B weights were staged on ds-serv6 under root-backed storage. A serving-capacity check estimated that the staged weights require four A5000 GPUs under our vLLM budget, while only one GPU was safely free in the live snapshot, so the reported generation and downstream results use the 9B fallback.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name par-

tials, and small typos.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B fallback-model prompts within context limits on larger schemas.

The deterministic Cypher layer borrows from production lessons in the BalkanID Cypher system: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k.

6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and

Gate	Evidence checked	Failure mode caught
Read-only safety	blocked Cypher write/admin tokens	destructive or operational queries
Schema validity	labels, relationship types, properties, categorical values	hallucinated graph vocabulary or values
Direction validity	observed relationship directions	reversed or invalid traversals
Cypher analysis and normalization	structure extraction, rewrite safety, RETURN DISTINCT	unsafe rewrites or duplicate/noisy rows
Question constraints	quoted values use exact literals	fuzzy matching of exact user values
Execution	query runs and returns rows	syntactic validity without answerability
LLM judge	ambiguity, semantic alignment, schema use	executable but semantically weak examples

Table 1: PIPE-Cypher candidate acceptance gates.

Graph	Target/category	Binding limit	Min seed capacity	All categories meet target
FinBench	250	300	300	Yes
SNB	125	200	200	Yes

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

Setting	Value
Accepted examples	3,000
Primary graph	LDBC FinBench, 2,000 examples
Secondary graph	LDBC SNB, 1,000 examples
Categories	8 balanced categories, 375 each
Generation/judge model	Local Qwen3.5-9B fallback
Execution backend	Neo4j Community, two live databases
Judge audit packet	80 sampled accepted/rejected pairs

Table 3: Full live experimental setup used for the generated benchmark artifact.

375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, but the Qwen3.5-9B fallback run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template concentration visible rather than hiding it.

The full-run failure taxonomy is concentrated in diversity/duplicate controls during recovery and empty execution results; only 2/4,777 candidates were schema-invalid after deterministic Cypher governance.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over both graphs and all eight categories. The calibration tooling tracks graph, category, difficulty, strategy, and judge-decision coverage, and reports

Graph	Candidates	Accepted	Acceptance	Categories at target
FinBench	3,404	2,000	0.588	8/8
SNB	1,373	1,000	0.728	8/8
Total	4,777	3,000	0.628	16/16

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

Export artifact	Examples	FinBench	SNB	Train	Dev	Test
Live full benchmark	3,000	2,000	1,000	2,408	296	296

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

agreement, precision/recall, specificity, balanced accuracy, and Cohen’s kappa once human labels are filled. The current draft treats these labels as pending human review rather than as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted

Artifact property	Value
Categories	8 balanced categories
FinBench/category	250
SNB/category	125
Difficulty split	1,521 easy / 1,479 medium
Unique labels / rel. types	7 / 9
Read/syntax/schema/exec/judge gates	3,000/3,000

Table 6: Distribution and gate summary for the exported full benchmark artifact.

Split	Examples	Parse	Schema	Exec. success	Exec. acc.	Answer F1
Live full test	296	0.959	0.905	0.622	0.189	0.189

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

Limitations

Execution validity does not guarantee semantic correctness. LLM judges can over-accept plausible but wrong examples, so the final study includes a small human audit for calibration. Generated artifacts may contain private schema details or sampled values, so organizations should apply normal data governance controls.

Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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Metric	Value
Question Distinct-1	0.041
Question Distinct-2	0.088
Question self-BLEU-2 (sampled)	0.826
Unique query-signature ratio	0.010
Top query-signature share	0.083
Category normalized entropy	1.000
Graph-category normalized entropy	0.980
Difficulty normalized entropy	1.000
Label coverage	0.412
Relationship-type coverage	0.375
Property-name coverage	0.407
Aggregation / negation / ordering rates	0.500 / 0.125 / 0.125

Table 8: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant.

Failure bucket	Count	Share of rejected
Diversity/duplicate control	1,335	0.751
Empty result	374	0.210
Judge semantic reject	66	0.037
Schema invalid	2	0.001

Table 9: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

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A Additional Results

B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

- Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark

Setting	Graph	Records	Accepted	Acceptance	Categories at target
Unconstrained LLM	FinBench	0	0	0.000	0/8
Unconstrained LLM	SNB	0	0	0.000	0/8
Reverse-only	FinBench	400	400	1.000	8/8
Reverse-only	SNB	402	400	0.995	8/8
Validators+repair	FinBench	400	400	1.000	8/8
Validators+repair	SNB	402	400	0.995	8/8
No retrieval	FinBench	429	400	0.932	8/8
No retrieval	SNB	402	400	0.995	8/8
No rewrite	FinBench	423	400	0.946	8/8
No rewrite	SNB	402	400	0.995	8/8
No LLM judge	FinBench	401	400	0.998	8/8
No LLM judge	SNB	406	400	0.985	8/8
Full PIPE-Cypher	FinBench	416	400	0.962	8/8
Full PIPE-Cypher	SNB	402	400	0.995	8/8

Table 10: Live target-50 ablation evidence with local Qwen3.5-9B. Each graph run targets 50 accepted examples per category.

Setting	Graph	Read-only	Syntax	Schema	Exec.	Judge
Unconstrained LLM	FinBench	0.000	0.000	0.000	0.000	0.000
Unconstrained LLM	SNB	0.000	0.000	0.000	0.000	0.000
Reverse-only	FinBench	1.000	1.000	1.000	1.000	1.000
Reverse-only	SNB	1.000	1.000	0.995	0.995	0.995
Validators+repair	FinBench	1.000	1.000	1.000	1.000	1.000
Validators+repair	SNB	1.000	1.000	0.995	0.995	0.995
No retrieval	FinBench	1.000	1.000	0.995	0.995	0.932
No retrieval	SNB	1.000	1.000	0.995	0.995	0.995
No rewrite	FinBench	1.000	1.000	1.000	1.000	0.946
No rewrite	SNB	1.000	1.000	0.995	0.995	0.995
No LLM judge	FinBench	1.000	1.000	1.000	1.000	0.998
No LLM judge	SNB	1.000	1.000	0.995	0.995	0.985
Full PIPE-Cypher	FinBench	1.000	1.000	1.000	1.000	0.962
Full PIPE-Cypher	SNB	1.000	1.000	0.995	0.995	0.995

Table 11: Quality-gate rates for the live target-50 ablation suite. Rates are computed over all generated records in each graph/setting.

over enterprise-style property graphs using local models.

Evidence. The full Qwen3.5-9B fallback run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

Key artifacts. benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

Status. Supported by live 9B fallback evidence; 35B target run is blocked by current GPU capacity.

Risk. Generation quality may change with the target 35B model or with private enterprise schemas beyond FinBench/SNB.

- Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

Evidence. The pipeline validates read-only safety, schema-visible labels/properties/relationships, observed relationship direction, categorical values, con-

Audit packet property	Value
Rows	80
Judge accept / reject	40 / 40
FinBench / SNB rows	48 / 32
Easy / medium rows	26 / 54
Labeled rows	0

Table 12: Post-hoc judge calibration packet coverage. Human labels are pending and are not used as a generation gate.

Metric	Point	95% CI	SE	N
Parse valid	0.959	[0.936, 0.980]	0.012	296
Schema valid	0.905	[0.872, 0.939]	0.017	296
Execution success	0.622	[0.564, 0.676]	0.028	296
Execution accuracy	0.189	[0.145, 0.233]	0.022	296
Answer F1	0.189	[0.149, 0.236]	0.023	296

Table 13: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

textual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

Key artifacts. code: validator.py; code: cypher_parser.py; code: question_constraints.py; snapshot: failure taxonomy; +1 more

Status. Supported by code, tests, and full-run failure taxonomy.

Risk. Semantic correctness still depends on execution evidence and judge review; human audit labels are pending.

- Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

Evidence. The target-50 FinBench/SNB ablation suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed the paper-readiness audit with logs, model IDs, code revision, run summaries, and gate-rate availability. Appendix-only tables and a figure report the audited target-50 evidence while the

target-100 follow-up and seeded target-50 repeat continue for stronger final reliability and variance claims.

Key artifacts. script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +8 more

Status. Supported by audited target-50 appendix evidence; target-100 and seeded target-50 repeat runs are still pending for stronger final reliability claims.

Risk. A single target-50 suite is the minimum publishable ablation scale, not the strongest possible evidence. If target-100 or repeated-seed runs complete, final claims should use those sensitivity results instead of relying only on target-50.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

Evidence. The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, and structural feature rates, making template concentration visible instead of hidden.

Key artifacts. snapshot: diversity report; paper table: diversity; paper figure: diversity diagnostics; paper figure: full export distribution

Status. Supported by full-export statistics and diversity diagnostics.

Risk. Low query-signature diversity in the 9B fallback run remains a limitation and ablation target.

- Claim 5.** The generated benchmark is useful for downstream Text2Cypher evaluation.

Evidence. A zero-shot local Qwen3.5-9B downstream evaluation over the 296-example full test split reports parse validity, schema validity, execution success, execution accuracy, answer F1, per-category breakdowns, and fixed-seed bootstrap uncertainty intervals. The model reaches 0.189 execution accuracy with a 95% bootstrap interval of [0.145, 0.233] and 0.622 execution success with interval [0.564, 0.676], showing the benchmark

442	can discriminate easy aggregation from harder	<i>Risk.</i> A later 35B run may change yield, diver-	489
443	operational categories.	sity, judge behavior, and downstream conclu-	490
444	<i>Key artifacts.</i> downstream summary; snap-	sions.	491
445	shot: downstream uncertainty; research note:	C Prompt Contracts	492
446	downstream evaluation; paper table: down-	PIPE-Cypher treats prompts as versioned imple-	493
447	stream evaluation; +3 more	mentation artifacts. The list summarizes the prompt	494
448	<i>Status.</i> Supported by live downstream evalua-	contracts used for generation, repair, judging, and	495
449	tion against FinBench/SNB databases, with	downstream evaluation; hashes fingerprint the full	496
450	bootstrap uncertainty computed from row-	prompt constants in the codebase.	497
451	level outputs.		
452	<i>Risk.</i> Only one downstream model has been	1. Template generation. Stage: Workload pro-	498
453	evaluated so far.	posal. SHA-256: 10b25b.	499
454	6. Claim 6. Automated LLM-judge review can	<i>Contract.</i> Schema-only labels, relationships,	500
455	replace human-in-the-loop generation gates	properties, and categorical values; Realistic	501
456	while still exposing judge risk.	enterprise analyst wording; At most two typed	502
457	<i>Evidence.</i> The pipeline uses deterministic val-	slots and JSON-only output	503
458	idation plus LLM-judge review as the gener-		
459	ation gate. A post-hoc 80-row judge audit	2. Reverse binding. Stage: Graph grounding.	504
460	packet preserves 40 judge accepts, 40 judge	SHA-256: 48e949.	505
461	rejects, both graphs, and all eight categories,	<i>Contract.</i>	506
462	and the analyzer reports agreement, preci-	Read-only	507
463	sion/recall, specificity, balanced accuracy, and	MATCH/WHERE/RETURN	508
464	Cohen’s kappa once human labels are filled.	DISTINCT/LIMIT only; Slot variables named	509
465	<i>Key artifacts.</i> code: judge.py; code: calibra-	exactly as requested; Forward relationship	510
466	tion.py; code: judge_audit_packet.py; snap-	directions from the schema	
467	shot: judge audit packet v2; +2 more		
468	<i>Status.</i> Pipeline gate is implemented; audit	3. Cypher generation. Stage: Candidate query.	511
469	packet coverage is supported.	SHA-256: 61c557.	512
470	<i>Risk.</i> Judge-human agreement is not yet	<i>Contract.</i> Only schema-visible constructs and	513
471	known because human labels are still blank.	observed directions; RETURN DISTINCT for	514
472	7. Claim 7. The intended 35B local genera-	set returns and exact equality for quoted val-	515
473	tion/judge path is staged but not currently	ues; Context columns, categorical hints, place-	516
474	runnable under observed shared-GPU con-	holderized retrieval, and no writes	517
475	straints.		
476	<i>Evidence.</i> Qwen3.5-35B-A3B is staged on ds-	4. Repair. Stage: Validation feedback. SHA-	518
477	serv6, but the capacity checker reports 68,573	256: 56c419.	519
478	MiB of weights, four required A5000 GPUs	<i>Contract.</i> Preserve question intent while fix-	520
479	under the conservative vLLM budget, and	ing validation or execution issues; Keep query	521
480	only one safe GPU in the latest snapshot.	read-only and schema-grounded; Return only	522
481	<i>Key artifacts.</i> script: check vllm capacity;	corrected Cypher	523
482	script: render vllm capacity snapshot; re-		
483	search note: model availability; research note:	5. LLM judge. Stage: Quality gate. SHA-256:	524
484	qwen35b capacity snapshot 20260601; +1	073938.	525
485	more	<i>Contract.</i> Inputs include question, Cypher,	526
486	<i>Status.</i> Documented blocker; current paper	schema slice, execution rows, and validation	527
487	results are explicitly reported as Qwen3.5-9B	summary; Strict JSON scores for ambiguity,	528
488	fallback evidence.	semantic alignment, schema use, and diffi-	529
		culty; Pass only useful, unambiguous enter-	530
		prise benchmark examples	531
		6. Downstream Text2Cypher. Stage: Model	532
		evaluation. SHA-256: 4c07ff.	533

534 *Contract.* Read-only Cypher only; Schema-
535 visible constructs and exact direction preser-
536 vation; RETURN DISTINCT, count/ranking
537 rules, and no explanations

538 D Representative Accepted Examples

539 The examples below are selected in stable identifier
540 order from the tracked full-export snapshot, one per
541 graph/category cell when available. They show the
542 natural-language question, accepted Cypher, struc-
543 tural tags, gate status, and a bounded execution-
544 result sample.

545 1. FinBench / Boolean Existence / easy. Ques- 546 tion: Does account '187743809466009406' 547 have any outgoing transfer?

```
548 MATCH (src:Account {accountId:  
549 '187743809466009406'})  
550 -[:TRANSFER_TO]->  
551 (:Account)  
552 RETURN DISTINCT COUNT(DISTINCT src)  
553 > 0  
554 AS HasOutgoingTransfer
```

555 *Structure:* single_hop, aggregation.

556 *Relationships:* TRANSFER_TO.

557 *Gates:* RO/Syn/Schema/Exec/Judge.

558 *Result sample:* {HasOutgoingTransfer: True};
559 observed rows: 1

560 2. FinBench / Complex Aggregation / medium. 561 Question: What is the total transferred amount 562 from accounts owned by person 'Gwar'?

```
563 MATCH (p:Person {personName:  
564 'Gwar'})  
565 -[:OWN_ACCOUNT]->  
566 (src:Account) -[:TRANSFER_TO]->  
567 (dst:Account)  
568 RETURN DISTINCT SUM(t.amount) AS  
569 TotalTransferredAmount
```

570 *Structure:* join_heavy, aggregation.

571 *Relationships:* OWN_ACCOUNT, TRANS-
572 FER_TO.

573 *Gates:* RO/Syn/Schema/Exec/Judge.

574 *Result sample:* {TotalTransferredAmount:
575 14972318.41}; observed rows: 1

576 3. FinBench / Complex Retrieval / easy. Ques- 577 tion: Which accounts received transfers from 578 accounts owned by person 'Zof'?

```
579 MATCH (p:Person {personName:  
580 'Zof'})  
581 -[:OWN_ACCOUNT]-> (src:Account)  
582 -[:TRANSFER_TO]-> (dst:Account)  
583 RETURN DISTINCT dst.accountId AS  
584 AccountId, dst.accountType AS  
585 AccountType, dst.isBlocked AS  
586 IsBlocked  
587 LIMIT 300
```

Structure: join_heavy, bounded_result. 588

Relationships: OWN_ACCOUNT, TRANS-
589 FER_TO. 590

Gates: RO/Syn/Schema/Exec/Judge. 591

Result sample: {AccountId:
592 4687402787162554854, AccountType:
593 credit card, IsBlocked: False}; observed rows:
594 3 595

596 4. FinBench / Negation Difference / medium.

Question: Which accounts owned by person
597 'Kant' have not sent any transfers? 598

```
599 MATCH (p:Person {personName:  
600 'Kant'})  
601 -[:OWN_ACCOUNT]-> (a:Account)  
602 WHERE NOT (a) -[:TRANSFER_TO]->  
603 (:Account)  
604 RETURN DISTINCT a.accountId AS  
605 AccountId, a.accountType AS  
606 AccountType,  
607 a.isBlocked AS IsBlocked  
608 LIMIT 300
```

Structure: join_heavy, negation, 609
610 bounded_result.

Relationships: OWN_ACCOUNT, TRANS-
611 FER_TO. 612

Gates: RO/Syn/Schema/Exec/Judge. 613

Result sample: {AccountId:
614 4739757132830738341, AccountType:
615 merchant account, IsBlocked: False};
616 observed rows: 1 617

618 5. FinBench / Path Temporal / medium. Ques- 619 tion: Which accounts can receive money 620 within two transfer hops from accounts owned 621 by person 'Sossamon'?

```
622 MATCH (p:Person {personName:  
623 'Sossamon'}) -[:OWN_ACCOUNT]->  
624 (src:Account) -[:TRANSFER_TO*1..2]->  
625 (dst:Account)  
626 RETURN DISTINCT dst.accountId AS  
627 AccountId, dst.accountType AS  
628 AccountType, dst.isBlocked AS  
629 IsBlocked  
630 LIMIT 300
```

Structure: join_heavy, path, bounded_result. 631

Relationships: OWN_ACCOUNT, TRANS-
632 FER_TO. 633

Gates: RO/Syn/Schema/Exec/Judge. 634

Result sample: {AccountId:
635 4687402787162554854, AccountType:
636 credit card, IsBlocked: False}; observed rows:
637 4 638

639 6. FinBench / Ranking Topk / medium. Ques- 640 tion: For accounts owned by person 'Barry',

641	which account sent the highest total transfer	9. SNB / Boolean Existence / medium. Ques-	697
642	amount?	tion: Does person with id 6597069766828	698
643		like any post?	699
644	<pre>MATCH (p:Person {personName:</pre>	<pre>MATCH (p:Person {id:</pre>	700
645	<pre>'Barry'})</pre>	<pre>6597069766828})</pre>	701
646	<pre>-[:OWN_ACCOUNT]-></pre>	<pre>OPTIONAL MATCH (p) -[:LIKES]-></pre>	702
647	<pre>(src:Account)-[t:TRANSFER_TO]-></pre>	<pre>(post:Post)</pre>	703
648	<pre>(:Account)</pre>	<pre>RETURN DISTINCT COUNT(post) > 0 AS</pre>	704
649	<pre>WITH src, SUM(t.amount) AS</pre>	<pre>LikesAnyPost</pre>	705
650	<pre>totalAmount</pre>		
651	<pre>RETURN DISTINCT src.accountId AS</pre>		
652	<pre>AccountId, src.accountType AS</pre>		
653	<pre>AccountType, src.isBlocked AS</pre>		
654	<pre>IsBlocked,</pre>		
655	<pre>totalAmount</pre>		
656	<pre>ORDER BY totalAmount DESC</pre>		
	<pre>LIMIT 1</pre>		
657	Structure: join_heavy, aggregation, or-	Structure: single_hop, aggregation, optional.	706
658	der_rank, bounded_result.	Relationships: LIKES.	707
659	Relationships: OWN_ACCOUNT, TRANS-	Gates: RO/Syn/Schema/Exec/Judge.	708
660	FER_TO.	Result sample: {LikesAnyPost: True}; ob-	709
661	Gates: RO/Syn/Schema/Exec/Judge.	erved rows: 1	710
662	Result sample: {AccountId:	10. SNB / Complex Aggregation / medium.	711
663	4732438783436260772, AccountType:	Question: How many distinct posts	712
664	internet account, IsBlocked: False}; observed	are in forums joined by person with id	713
665	rows: 1	6597069766845?	714
666	7. FinBench / Simple Aggregation / easy. Ques-	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>	715
667	tion: How many accounts are owned by per-	<pre>(p:Person {id: 6597069766845})</pre>	716
668	son 'Kaewsuktae'?	<pre>MATCH (forum) -[:CONTAINER_OF]-></pre>	717
669		<pre>(post:Post)</pre>	718
670	<pre>MATCH (p:Person {personName:</pre>	<pre>RETURN DISTINCT COUNT(DISTINCT post)</pre>	719
671	<pre>'Kaewsuktae'}) -[:OWN_ACCOUNT]-></pre>	<pre>AS</pre>	720
672	<pre>(a:Account)</pre>	<pre>JoinedForumPostCount</pre>	721
673	<pre>RETURN DISTINCT COUNT(DISTINCT a) AS</pre>		
	<pre>AccountCount</pre>		
674	Structure: single_hop, aggregation.	Structure: join_heavy, aggregation.	722
675	Relationships: OWN_ACCOUNT.	Relationships: CONTAINER_OF,	723
676	Gates: RO/Syn/Schema/Exec/Judge.	HAS_MEMBER.	724
677	Result sample: {AccountCount: 1}; observed	Gates: RO/Syn/Schema/Exec/Judge.	725
678	rows: 1	Result sample: {JoinedForumPostCount: 1};	726
679	8. FinBench / Simple Retrieval / easy. Ques-	observed rows: 1	727
680	tion: Which accounts are owned by person	11. SNB / Complex Retrieval / easy. Question:	728
681	'Barry'?	Which people are members of forums contain-	729
682		ing posts tagged 'Manuel_Noriega'?	730
683	<pre>MATCH (p:Person {personName:</pre>	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-></pre>	731
684	<pre>'Barry')</pre>	<pre>(p:Person), (forum)</pre>	732
685	<pre>-[:OWN_ACCOUNT]-> (a:Account)</pre>	<pre>-[:CONTAINER_OF]-></pre>	733
686	<pre>RETURN DISTINCT a.accountId AS</pre>	<pre>(post:Post) -[:HAS_TAG]-> (tag:Tag</pre>	734
687	<pre>AccountId, a.accountType AS</pre>	<pre>{name: 'Manuel_Noriega'})</pre>	735
688	<pre>AccountType,</pre>	<pre>RETURN DISTINCT p.id AS PersonId</pre>	736
689	<pre>a.isBlocked AS IsBlocked</pre>	<pre>LIMIT 200</pre>	737
	<pre>LIMIT 300</pre>		
690	Structure: single_hop, bounded_result.	Structure: join_heavy, bounded_result.	738
691	Relationships: OWN_ACCOUNT.	Relationships: CONTAINER_OF,	739
692	Gates: RO/Syn/Schema/Exec/Judge.	HAS_MEMBER, HAS_TAG.	740
693	Result sample: {AccountId:	Gates: RO/Syn/Schema/Exec/Judge.	741
694	4732438783436260772, AccountType:	Result sample: {PersonId: 4398046511220};	742
695	internet account, IsBlocked: False}; observed	observed rows: 19	743
696	rows: 1	12. SNB / Negation Difference / medium. Ques-	744
		tion: Which members of forum 'Wall of Celso	745
		Oliveira' have not liked any post?	746

```

MATCH (forum:Forum {title: 'Wall of
Celso Oliveira'}) -[:HAS_MEMBER]->
(p:Person)
WHERE NOT (p) -[:LIKES]-> (:Post)
RETURN DISTINCT p.id AS PersonId,
p.firstName AS FirstName, p.lastName
AS
LastName
LIMIT 200

```

Structure: join_heavy, negation, bounded_result.

Relationships: HAS_MEMBER, LIKES.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {FirstName: K., LastName: Sen, PersonId: 94}; observed rows: 1

13. **SNB / Path Temporal / easy.** *Question:* Which people are within two knows hops of person with id 4398046511136?

```

MATCH (src:Person {id:
4398046511136})
-[:KNOWS*1..2]-> (dst:Person)
RETURN DISTINCT dst.id AS PersonId,
dst.firstName AS FirstName,
dst.lastName
AS LastName
LIMIT 200

```

Structure: single_hop, path, bounded_result.

Relationships: KNOWS.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {FirstName: Rafael, LastName: Fernández, PersonId: 4398046511333}; observed rows: 25

14. **SNB / Ranking Topk / medium.** *Question:* Among forums tagged 'James_K._Polk', which forum has the most members?

```

MATCH (forum:Forum) -[:HAS_TAG]->
(tag:Tag {name: 'James_K._Polk'})
MATCH (forum) -[:HAS_MEMBER]->
(person:Person)
WITH forum, COUNT(DISTINCT person)
AS
memberCount
RETURN DISTINCT forum.title AS
ForumTitle, memberCount
ORDER BY memberCount DESC
LIMIT 1

```

Structure: join_heavy, aggregation, order_rank, bounded_result.

Relationships: HAS_MEMBER, HAS_TAG.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {ForumTitle: Wall of Javed Khan, memberCount: 33}; observed rows: 1

15. **SNB / Simple Aggregation / easy.** *Question:* How many posts are tagged 'Vietnam'?

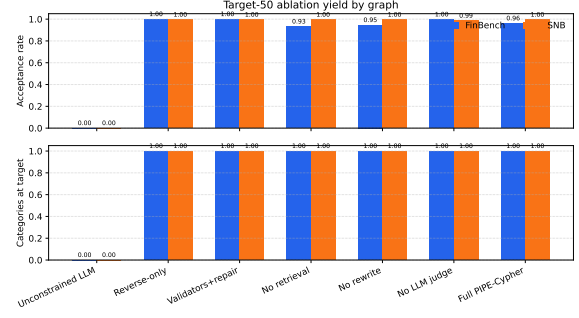


Figure 1: Audited target-50 ablation suite over FinBench and SNB. The suite reaches all graph/variant/category targets and is included as appendix evidence while the larger target-100 and seeded repeat suites continue running for stronger final reliability evidence.

```

MATCH (post:Post) -[:HAS_TAG]->
(tag:Tag
{name: 'Vietnam'})
RETURN DISTINCT COUNT(DISTINCT post)
AS
PostCount

```

Structure: single_hop, aggregation.

Relationships: HAS_TAG.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {PostCount: 1}; observed rows: 1

16. **SNB / Simple Retrieval / easy.** *Question:* Which post IDs did person with id 4398046511124 like?

```

MATCH (p:Person {id:
4398046511124})
-[:LIKES]-> (post:Post)
RETURN DISTINCT post.id AS PostId
LIMIT 200

```

Structure: single_hop, bounded_result.

Relationships: LIKES.

Gates: RO/Syn/Schema/Exec/Judge.

Result sample: {PostId: 343597385744}; observed rows: 5

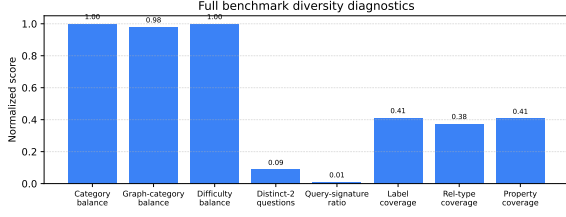


Figure 2: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the fallback seeded-template run.

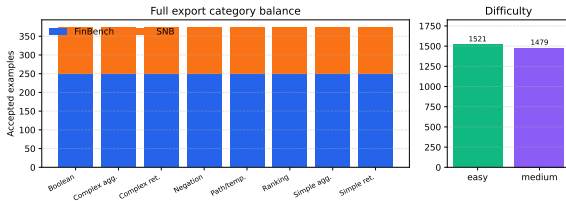


Figure 3: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

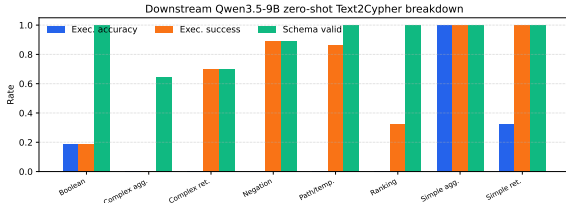


Figure 4: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

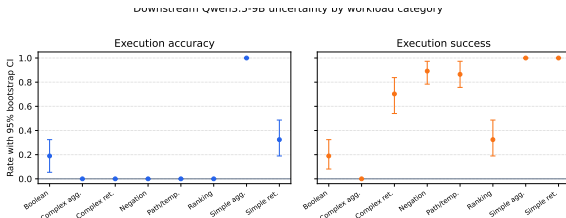


Figure 5: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.